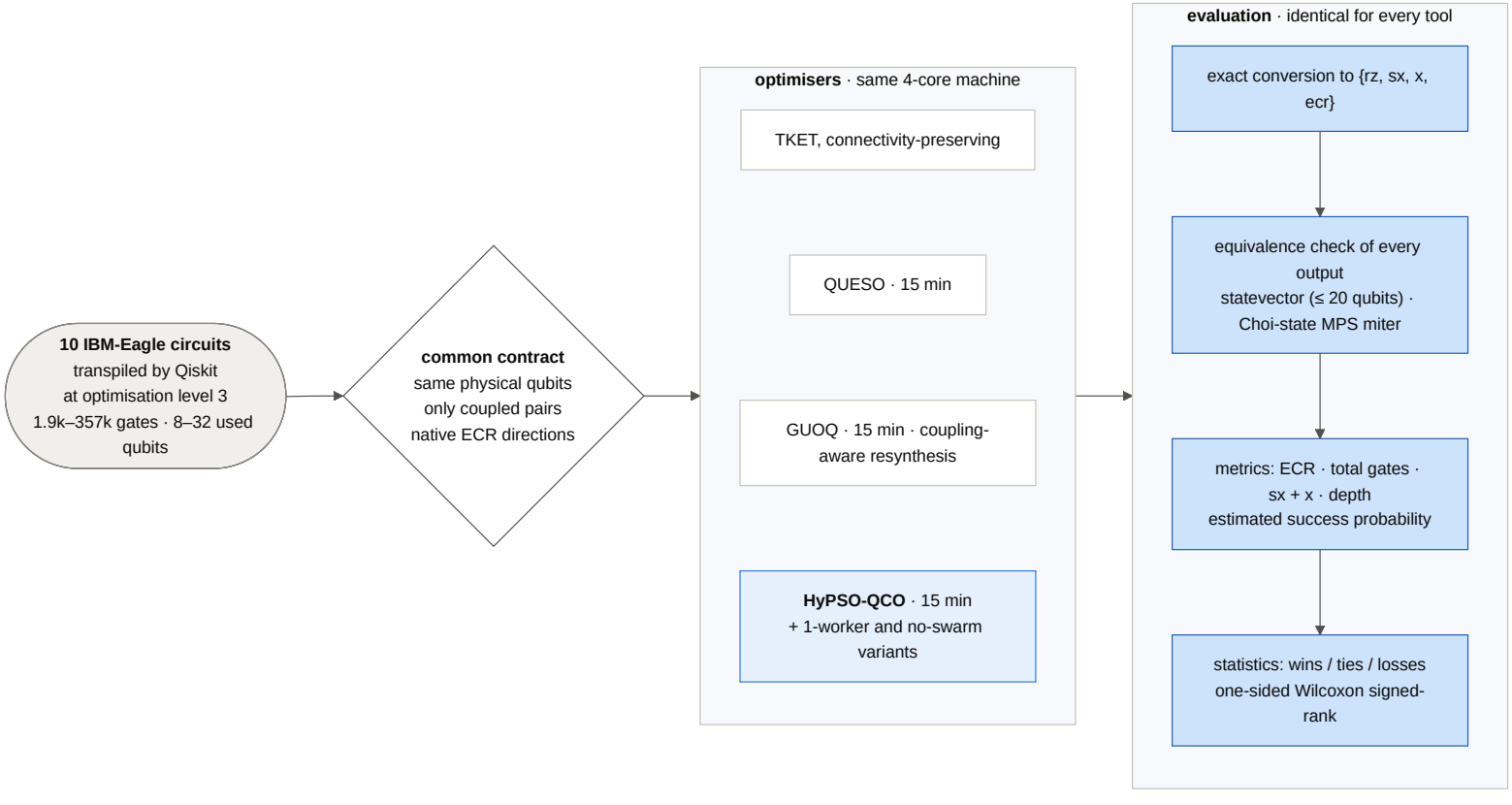
10 IBM-Eagle circuits
transpiled by Qiskit
at optimisation level 3
1.9k–357k gates · 8–32 used qubits
common contract
same physical qubits
only coupled pairs
native ECR directions
optimisers · same 4-core machine
TKET, connectivity-preserving
QUESO · 15 min
GUOQ · 15 min · coupling-aware resynthesis
HyPSO-QCO · 15 min
+ 1-worker and no-swarm variants
evaluation · identical for every tool
exact conversion to {rz, sx, x, ecr}
equivalence check of every output
statevector (≤ 20 qubits) · Choi-state MPS miter
metrics: ECR · total gates · sx + x · depth
estimated success probability
statistics: wins / ties / losses
one-sided Wilcoxon signed-rank